

Supplementary materials

Abstract

To accompany Townend, J., et al., 2012. Three-dimensional variations in present-day tectonic stress along the Australia-Pacific plate boundary in New Zealand. *Earth and Planetary Science Letters*, <http://dx.doi.org/10.1016/j.epsl.2012.08.003>.

1. Comparison of hypocenters and focal mechanisms

A comparison of the hypocenters obtained using NonLinLoc and the 3D seismic velocity model with those obtained by GeoNet and a 1D seismic velocity model is illustrated in Figure S1. The epicenters and depths differ, on average, by less than 5 km and 10 km, respectively, and reveal some systematic differences in hypocenters: in particular, the NonLinLoc hypocenters tend to have systematically greater depths than the GeoNet ones, due to the incorporation of high velocities corresponding to the Pacific slab, in particular, in the 3D velocity model used. These discrepancies are small in comparison to the scales at which we cluster focal mechanisms for stress parameter estimation.

Figure S2 compares the focal mechanisms obtained in this study with regional moment tensor solutions (Ristau, 2008). To quantify the degree of similarity between two different estimates of a particular focal mechanism, we compute the minimum rotation required to align the estimates (e.g. Arnold and Townend, 2007):

$$a = \cos^{-1} \left(\frac{\text{tr}(\mathbf{R}_1^T \mathbf{R}_2) - 1}{2} \right) \quad (1)$$

where $\mathbf{R}_1$ and $\mathbf{R}_2$ are rotation matrices describing each focal mechanism with respect to geographic coordinates. In general, the three estimates of each of the 30 focal mechanisms illustrated in Figure S2 are qualitatively similar, although some clear discrepancies exist. The two estimates obtained using the NonLinLoc locations (NonLinLoc+probabilistic, NonLinLoc+Fpfit) are typically more similar than either is with the GeoNet+CMT solution, as might be expected given the differences between the NonLinLoc and GeoNet hypocenters illustrated in Figure S1.

2. Tables of stress estimation results for $k=100$ and $k=41$

Tables S1 and S2 list the cluster centroid locations and stress parameter estimates for the $k = 100$ analysis described in the manuscript and $k = 41$ analysis, for comparison. Tables S3 and S4 summarize the stress results obtained for clusters in the vicinity of Christchurch using focal mechanisms recorded before and after the M_W 6.2 Christchurch earthquake of 22 February 2011.

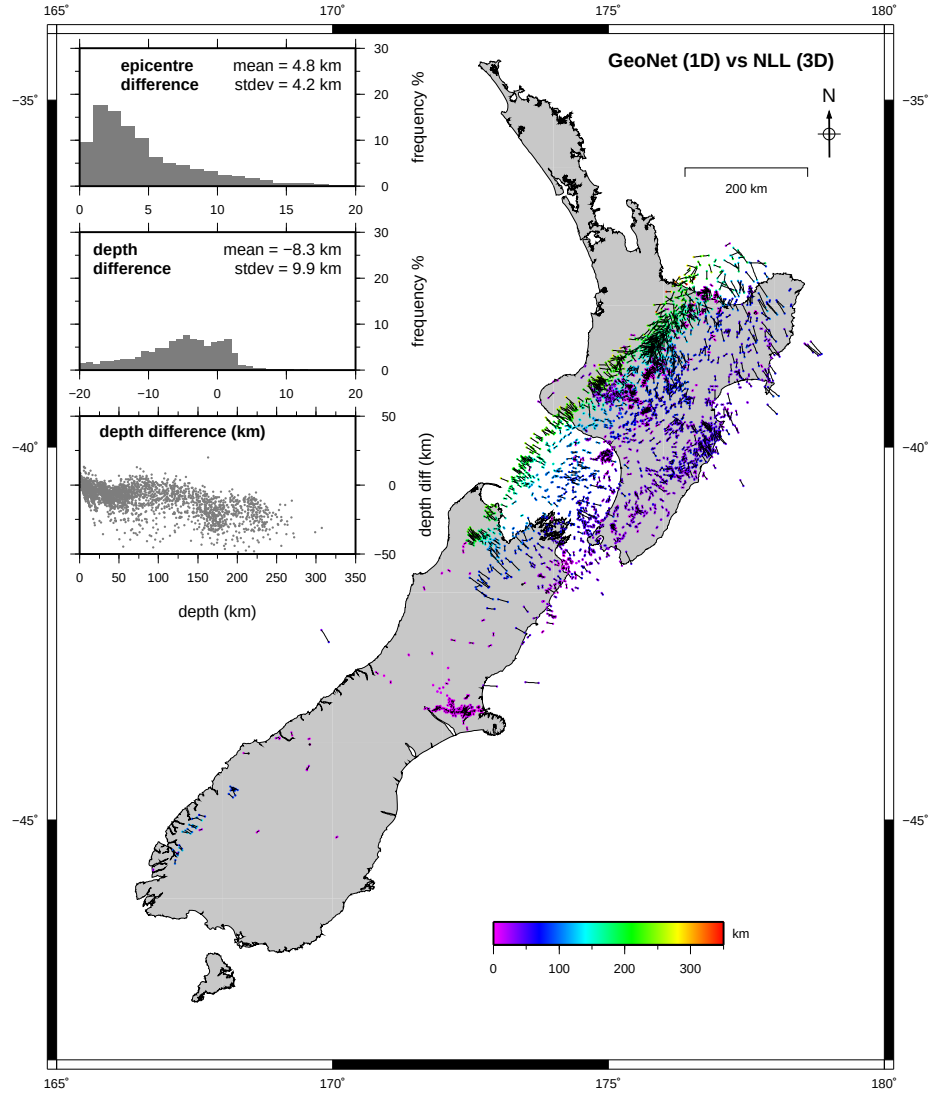


Figure S1: A comparison of the epicenters obtained by GeoNet using a 1D velocity model and in this study using a 3D velocity model and NonLinLoc. The map illustrates the differences in epicenters (with colored lines linking the two epicenters for each earthquake) and the insets summarize the horizontal and vertical differences.

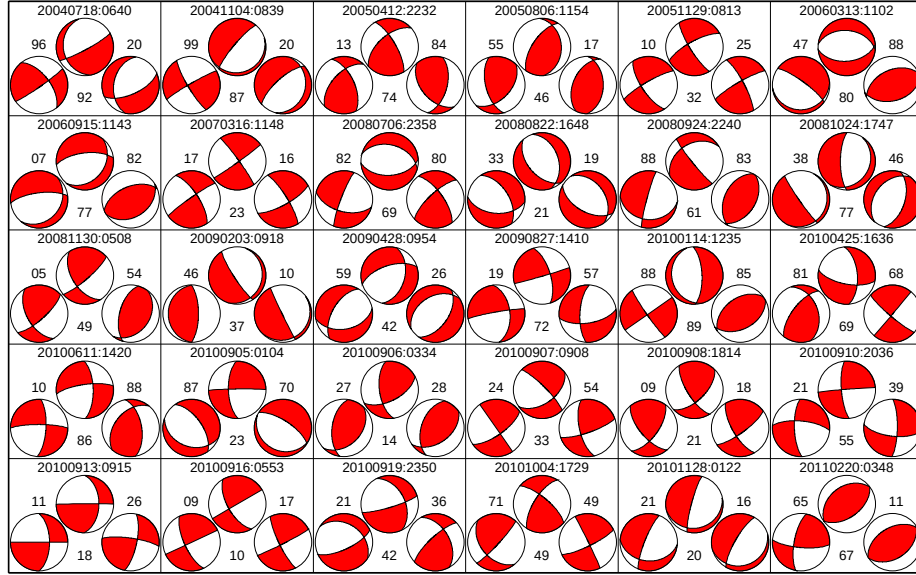


Figure S2: A comparison of 30 focal mechanisms obtained using the NonLinLoc hypocenters and Walsh et al. (2009) focal mechanism estimation algorithm (upper mechanism in each panel) with the results obtained using the NonLinLoc hypocenters and Fpfit (lower left), and using GeoNet's hypocenters and a regional centroid moment tensor code (lower right; Ristau, 2008). The top number in each panel gives the origin date and time of the event and the figure between each pair of focal mechanisms specifies the minimum rotation required to align the two mechanisms.

3. Focal mechanism clustering and stress maps for $k=41$

Figures S3–S6 illustrate the focal mechanism clusters obtained for $k = 41$ and the corresponding stress results. Figure S7 shows the differences in estimated S_{Hmax} orientation before and after the M_W 6.2 Christchurch earthquake of 22 February 2011.

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Table S1: Summary of stress results for $k = 100$ clusters. S_{Hmax} denotes the axis of maximum horizontal compressive stress computed from Euler angles ϕ_{GS} , θ_{GS} and ψ_{GS} specifying the orientations of the principal stresses and the stress ratio $v = (S_1 - S_2)/(S_1 - S_3)$ using methods described by Arnold and Townend (2007) and Lund and Townend (2007). Subscripts G and S denote geographic and principal stress coordinate systems, respectively.

Cluster	Cluster centroid			S_{Hmax} azimuth ($^\circ$)			Stress ratio (v)			Stress orientation parameters			S_v
	Long. ($^\circ$)	Lat. ($^\circ$)	Depth (km)	10% limit	Mean	90% limit	10% limit	Mean	90% limit	ϕ_{GS}	θ_{GS}	ψ_{GS}	
1	178.3	-38.7	46	37	75	109	0.44	0.78	0.98	82.1	140.0	307.8	S_1
2	175.1	-41.3	33	40	57	74	-0.04	0.19	0.43	266.9	131.1	137.1	S_1
3	175.7	-39.0	8	54	66	79	0.55	0.62	0.72	185.5	176.3	29.0	S_1
4	176.4	-39.3	41	16	47	77	0.08	0.45	0.84	214.0	51.0	120.0	S_2
5	171.9	-43.5	7	123	135	146	0.38	0.55	0.72	315.1	94.1	96.0	S_2
6	175.9	-39.5	65	75	92	108	0.12	0.35	0.57	289.9	125.4	126.2	S_2
7	172.4	-43.6	7	100	111	122	0.42	0.53	0.63	290.9	92.6	90.0	S_2
8	174.5	-41.0	68	39	59	78	0.12	0.29	0.49	85.8	41.2	231.7	S_1
9	175.0	-40.1	24	61	74	86	0.31	0.47	0.62	249.4	106.9	60.5	S_2
10	177.6	-39.0	51	30	43	56	0.30	0.54	0.78	44.7	78.2	251.9	S_2
11	174.2	-41.6	17	96	109	121	0.33	0.51	0.70	288.5	96.6	88.2	S_2
12	175.9	-38.9	9	46	61	75	0.28	0.43	0.57	157.8	167.9	6.6	S_1
13	176.0	-38.2	225	59	87	116	0.03	0.25	0.51	227.8	57.7	340.0	S_3
14	176.1	-38.6	10	33	48	63	0.19	0.49	0.76	48.5	102.2	273.0	S_2
15	176.3	-40.4	35	60	78	94	0.46	0.69	0.91	81.2	51.9	256.6	S_2
16	173.9	-41.9	24	84	97	110	0.45	0.64	0.83	277.1	111.7	87.7	S_2
17	173.4	-40.5	177	27	42	57	0.34	0.53	0.73	39.4	96.9	216.6	S_3
18	174.6	-41.6	42	54	67	79	0.74	0.88	0.98	66.8	99.4	248.3	S_2
19	175.1	-39.8	78	144	173	202	0.20	0.48	0.78	186.1	59.3	42.3	S_2
20	176.4	-37.7	229	64	84	104	0.07	0.26	0.44	335.0	142.9	166.5	S_1
21	175.0	-40.6	48	61	80	98	0.28	0.58	0.85	261.9	102.8	118.5	S_2
22	177.3	-37.6	173	121	146	175	0.26	0.65	0.95	143.4	76.1	165.7	S_3
23	175.1	-41.1	35	51	64	77	0.31	0.48	0.64	269.9	149.0	126.4	S_1
24	172.5	-43.5	13	100	119	136	0.40	0.74	0.98	117.6	97.5	63.7	S_2

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Cluster	Cluster centroid			S_{Hmax} azimuth			Stress ratio (ν)			Stress orientation parameters			S_ν
	Long. (°)	Lat. (°)	Depth (km)	10% limit	Mean	90% limit	10% limit	Mean	90% limit	ϕ_{GS}	θ_{GS}	ψ_{GS}	
25	175.8	-38.8	133	69	84	99	0.05	0.22	0.40	48.4	129.7	214.8	S_1
26	176.8	-37.8	9	38	53	67	0.42	0.53	0.64	279.0	12.2	41.6	S_1
27	172.5	-42.7	11	102	126	150	0.23	0.61	0.95	307.2	80.8	68.2	S_2
28	174.3	-39.8	136	64	90	116	0.31	0.53	0.74	56.4	139.3	209.2	S_1
29	176.5	-38.1	174	74	90	105	0.00	0.15	0.32	101.3	78.2	217.0	S_3
30	170.8	-43.0	15	90	118	146	0.04	0.45	0.89	116.9	95.2	259.6	S_2
31	177.6	-38.5	68	37	51	65	0.08	0.34	0.59	50.7	91.1	242.8	S_2
32	172.7	-43.5	6	107	126	146	0.21	0.52	0.82	119.6	109.4	50.3	S_2
33	174.3	-40.4	104	101	116	130	0.53	0.68	0.82	305.0	125.3	140.5	S_3
34	172.9	-40.9	208	87	123	178	0.29	0.54	0.77	126.9	67.0	196.7	S_3
35	177.6	-38.1	7	57	76	95	0.23	0.70	0.98	256.1	91.1	85.8	S_2
36	175.2	-39.2	151	57	85	115	0.20	0.40	0.59	46.0	132.3	203.5	S_3
37	174.6	-40.6	83	72	92	112	0.27	0.53	0.79	274.2	103.5	118.4	S_2
38	176.4	-37.7	313	37	82	123	0.00	0.33	0.73	358.3	141.3	182.4	S_1
39	175.7	-38.7	171	68	85	102	0.05	0.16	0.29	256.0	81.9	335.1	S_3
40	174.3	-39.4	258	35	74	114	0.24	0.52	0.80	37.4	137.4	202.6	S_1
41	172.6	-41.2	207	-18	11	40	-0.07	0.09	0.25	341.1	103.5	199.7	S_3
42	169.3	-44.3	7	100	119	139	0.47	0.78	0.98	300.0	87.4	124.2	S_2
43	174.7	-39.7	120	73	89	105	0.15	0.35	0.56	258.9	72.0	319.0	S_3
44	178.0	-37.9	99	-8	27	68	0.09	0.31	0.54	66.4	54.9	201.3	S_3
45	176.6	-39.9	34	54	72	90	0.07	0.30	0.55	258.4	123.4	102.7	S_2
46	175.5	-39.0	124	96	144	185	0.14	0.70	0.98	115.0	149.3	235.4	S_1
47	176.5	-38.1	7	22	46	66	0.29	0.48	0.67	262.0	24.3	47.0	S_1
48	176.5	-40.2	59	47	60	73	-0.03	0.10	0.24	98.6	16.0	230.2	S_1
49	173.8	-40.7	117	56	125	195	0.04	0.20	0.39	124.8	70.4	180.6	S_3
50	175.8	-40.7	32	29	55	85	0.33	0.58	0.83	274.7	156.7	141.2	S_1

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Cluster	Cluster centroid			S_{Hmax} azimuth			Stress ratio (ν)			Stress orientation parameters			S_ν
	Long. (°)	Lat. (°)	Depth (km)	10% limit	Mean	90% limit	10% limit	Mean	90% limit	ϕ_{GS}	θ_{GS}	ψ_{GS}	
77	176.2	-38.6	129	1	32	66	0.25	0.53	0.80	11.3	126.2	207.5	S_3
78	174.8	-39.1	236	65	79	92	0.29	0.44	0.58	69.8	109.1	205.2	S_3
79	175.7	-39.5	19	9	20	31	0.62	0.75	0.88	201.3	64.0	79.9	S_2
80	173.0	-41.3	131	3	34	61	0.06	0.24	0.44	185.5	67.1	159.8	S_3
81	176.3	-39.7	52	59	72	84	-0.06	0.10	0.28	249.7	82.0	109.7	S_2
82	175.7	-39.7	24	21	36	50	-0.04	0.20	0.48	37.4	104.7	279.1	S_2
83	175.5	-38.6	190	36	82	124	0.06	0.23	0.43	3.5	124.7	183.2	S_3
84	176.0	-40.0	41	62	82	104	-0.03	0.18	0.43	315.3	158.5	145.6	S_2
85	176.2	-38.9	102	39	60	81	0.32	0.50	0.68	227.7	61.1	340.8	S_3
86	176.5	-40.3	25	117	149	183	0.07	0.38	0.70	130.0	60.4	139.9	S_3
87	176.8	-38.8	69	42	62	82	0.07	0.23	0.43	74.3	71.1	227.2	S_2
88	167.6	-45.1	91	107	139	181	0.06	0.33	0.62	313.4	96.1	359.9	S_3
89	175.3	-40.4	77	52	76	98	0.20	0.50	0.80	71.2	107.5	242.6	S_2
90	175.1	-39.5	111	65	93	121	0.24	0.55	0.88	66.5	147.5	229.5	S_1
91	176.1	-40.3	59	29	61	84	0.07	0.30	0.72	78.0	33.2	251.2	S_1
92	175.2	-38.8	241	84	102	120	0.13	0.31	0.49	259.7	62.1	328.9	S_3
93	177.0	-39.4	43	26	41	55	0.18	0.34	0.52	219.4	83.3	101.7	S_2
94	175.3	-39.9	30	51	72	93	0.27	0.54	0.82	67.0	128.1	254.2	S_2
95	176.6	-38.4	6	35	55	74	-0.02	0.23	0.57	234.0	114.1	88.6	S_2
96	173.7	-39.9	237	55	124	193	-0.01	0.24	0.53	229.2	60.7	357.5	S_3
97	172.4	-43.6	7	106	117	128	-0.05	0.11	0.26	297.3	84.9	89.9	S_2
98	175.7	-39.3	90	78	98	120	0.41	0.60	0.79	74.2	148.7	232.4	S_1
99	176.1	-38.3	188	67	87	106	-0.12	0.05	0.16	12.1	119.7	187.0	S_3
100	173.4	-40.2	209	92	137	184	0.18	0.35	0.53	179.4	52.5	14.7	S_3

Table S2: Summary of stress results for $k = 41$ clusters. S_{Hmax} denotes the axis of maximum horizontal compressive stress computed from Euler angles ϕ_{GS} , θ_{GS} and ψ_{GS} specifying the orientations of the principal stresses and the stress ratio $\nu = (S_1 - S_2)/(S_1 - S_3)$ using methods described by Arnold and Townend (2007) and Lund and Townend (2007). Subscripts G and S denote geographic and principal stress coordinate systems, respectively.

Cluster	Cluster centroid			S_{Hmax} azimuth ($^{\circ}$)			Stress ratio (ν)			Stress orientation parameters			S_{ν}
	Long. ($^{\circ}$)	Lat. ($^{\circ}$)	Depth (km)	10% limit	Mean	90% limit	10% limit	Mean	90% limit	ϕ_{GS}	θ_{GS}	ψ_{GS}	
1	175.2	-38.8	235	74	91	108	0.22	0.41	0.61	70.4	121.6	208.3	S_3
2	174.8	-39.1	235	68	82	96	0.28	0.43	0.56	251.4	69.5	332.2	S_3
3	173.2	-41.4	106	41	60	80	0.13	0.29	0.49	20.9	122.3	205.9	S_3
4	176.2	-40.4	43	51	63	74	0.24	0.32	0.42	88.7	32.7	236.7	S_1
5	175.8	-38.5	183	60	76	91	0.06	0.13	0.22	218.9	67.2	342.9	S_3
6	177.1	-37.7	172	100	130	160	0.19	0.43	0.64	138.3	68.1	188.4	S_3
7	174.1	-41.8	26	84	95	105	0.57	0.68	0.80	274.7	100.8	90.2	S_2
8	175.0	-41.2	35	52	64	76	0.34	0.44	0.54	88.4	36.5	232.4	S_1
9	176.1	-39.2	76	79	90	101	0.50	0.61	0.71	90.1	91.4	233.7	S_2
10	175.5	-40.8	36	53	66	79	0.35	0.48	0.61	92.9	37.3	227.3	S_1
11	174.1	-39.6	231	57	82	107	0.18	0.36	0.54	218.5	48.2	338.6	S_3
12	176.8	-37.8	9	39	53	67	0.38	0.50	0.62	277.3	12.1	43.9	S_1
13	176.0	-38.8	126	48	74	105	0.26	0.37	0.50	30.5	129.7	195.2	S_3
14	172.0	-43.5	7	107	118	129	0.48	0.59	0.71	298.2	93.9	97.1	S_2
15	177.7	-38.0	74	42	62	84	0.38	0.54	0.71	69.0	64.7	200.6	S_3
16	172.8	-42.0	48	73	91	108	0.07	0.34	0.66	86.0	103.3	246.3	S_2
17	176.9	-39.6	45	22	33	45	0.16	0.26	0.38	211.1	80.6	108.1	S_2
18	176.0	-38.8	9	45	59	72	0.15	0.30	0.45	213.2	167.3	63.3	S_1
19	169.4	-44.0	12	98	118	138	0.46	0.78	0.98	298.7	92.7	103.1	S_3
20	177.7	-38.9	54	33	45	57	0.26	0.42	0.58	46.1	84.6	250.7	S_2
21	175.7	-39.5	20	11	22	33	0.60	0.72	0.84	203.8	64.4	79.0	S_2
22	174.4	-41.1	60	48	64	78	0.25	0.42	0.58	75.7	58.4	237.1	S_2
23	176.2	-39.7	52	68	80	92	0.02	0.16	0.32	259.6	86.4	109.1	S_2
24	176.7	-40.0	51	54	67	79	0.18	0.29	0.41	44.6	170.4	247.2	S_1

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Cluster	Cluster centroid			S_{Hmax} azimuth			Stress ratio (ν)			Stress orientation parameters			S_ν
	Long. (°)	Lat. (°)	Depth (km)	10% limit	Mean	90% limit	10% limit	Mean	90% limit	ϕ_{GS}	θ_{GS}	ψ_{GS}	
25	173.5	-40.3	190	20	49	72	0.25	0.36	0.50	32.0	117.0	194.0	S_3
26	175.3	-39.2	135	78	97	116	0.26	0.41	0.55	58.5	134.4	206.8	S_1
27	174.6	-40.4	90	98	111	123	0.50	0.66	0.82	113.3	77.1	206.0	S_3
28	175.4	-39.3	16	78	90	101	0.29	0.39	0.49	157.3	166.9	338.9	S_1
29	176.3	-38.2	183	71	83	96	-0.16	0.04	0.13	70.1	100.5	216.4	S_3
30	175.7	-39.0	8	55	67	79	0.48	0.58	0.64	333.7	5.0	3.4	S_1
31	172.7	-41.1	197	117	173	215	0.10	0.24	0.39	317.6	114.7	190.8	S_3
32	175.4	-39.6	77	91	127	160	0.38	0.54	0.70	343.3	138.2	159.0	S_1
33	177.0	-38.4	76	55	69	82	0.10	0.22	0.34	89.4	60.4	224.2	S_3
34	167.7	-45.1	88	108	136	167	0.11	0.39	0.66	313.1	93.9	356.8	S_3
35	174.9	-39.2	33	46	57	67	0.16	0.27	0.40	237.2	64.3	89.9	S_2
36	173.9	-40.5	113	79	96	112	0.19	0.35	0.52	112.2	65.7	215.3	S_3
37	174.5	-39.7	132	73	87	100	0.29	0.44	0.60	65.6	130.5	224.3	S_1
38	175.0	-40.1	27	66	77	88	0.34	0.45	0.57	256.6	93.3	70.4	S_2
39	176.3	-37.7	253	59	81	102	0.15	0.31	0.48	354.0	141.5	181.7	S_1
40	176.5	-38.2	7	36	50	64	0.15	0.28	0.43	85.7	160.9	308.3	S_1
41	172.4	-43.6	7	105	115	125	0.37	0.45	0.53	115.2	91.9	270.0	S_2

Table S3: Summary of stress results before and after the M_W 6.2 Christchurch earthquake of 22 February 2011 for seven of $k = 100$ clusters. S_{Hmax} denotes the axis of maximum horizontal compressive stress computed from Euler angles ϕ_{GS} , θ_{GS} and ψ_{GS} specifying the orientations of the principal stresses and the stress ratio $v = (S_1 - S_2)/(S_1 - S_3)$ using methods described by Arnold and Townend (2007) and Lund and Townend (2007). Subscripts G and S denote geographic and principal stress coordinate systems, respectively.

Cluster	Cluster centroid			S_{Hmax} azimuth ($^\circ$)			Stress ratio (v)			Stress orientation parameters			S_v
	Long. (deg)	Lat. ($^\circ$)	Depth (km)	10% limit	Mean	90% limit	10% limit	Mean	90% limit	ϕ_{GS}	θ_{GS}	ψ_{GS}	
5 (early)	171.9	-43.5	7	123	135	146	0.38	0.55	0.72	315.1	94.1	96.0	S_2
5 (late)	171.9	-43.5	7	54	118	182	0.07	0.52	0.95	304.3	155.0	92.9	S_2
7 (early)	172.4	-43.6	7	100	111	122	0.42	0.53	0.63	290.9	92.6	90.0	S_2
7 (late)	172.4	-43.6	7	83	112	139	0.18	0.64	0.97	294.2	93.3	134.2	S_2
24 (early)	172.5	-43.5	12	101	118	135	0.54	0.80	0.98	117.4	95.3	237.0	S_2
24 (late)	172.5	-43.5	12	70	123	177	0.05	0.48	0.93	302.3	96.1	89.7	S_2
32 (early)	172.7	-43.6	5	110	127	143	0.26	0.54	0.81	122.9	105.8	239.7	S_2
32 (late)	172.7	-43.6	5	109	120	131	0.49	0.61	0.72	296.9	69.2	292.3	S_2
52 (early)	172.5	-43.6	5	109	120	131	0.23	0.38	0.53	120.3	94.8	269.9	S_2
52 (late)	172.5	-43.6	5	104	118	131	0.43	0.66	0.90	117.3	110.1	265.2	S_2
74 (early)	172.1	-43.5	7	98	109	120	0.48	0.60	0.72	289.2	93.8	94.0	S_2
74 (late)	172.1	-43.5	7	108	128	147	0.12	0.60	0.97	308.2	87.0	87.0	S_2
97 (early)	172.4	-43.6	7	106	117	128	-0.05	0.11	0.26	297.3	84.9	89.9	S_2
97 (late)	172.4	-43.6	7	100	124	147	0.07	0.55	0.96	304.5	107.4	96.0	S_2

Table S4: Summary of stress results before and after the M_W 6.2 Christchurch earthquake of 22 February 2011 for two of $k = 41$ clusters. S_{Hmax} denotes the axis of maximum horizontal compressive stress computed from Euler angles ϕ_{GS} , θ_{GS} and ψ_{GS} specifying the orientations of the principal stresses and the stress ratio $v = (S_1 - S_2)/(S_1 - S_3)$ using methods described by Arnold and Townend (2007) and Lund and Townend (2007). Subscripts G and S denote geographic and principal stress coordinate systems, respectively.

Cluster	Cluster centroid			S_{Hmax} azimuth ($^\circ$)			Stress ratio (v)			Stress orientation parameters			S_v
	Long. (deg)	Lat. ($^\circ$)	Depth (km)	10% limit	Mean	90% limit	10% limit	Mean	90% limit	ϕ_{GS}	θ_{GS}	ψ_{GS}	
14 (early)	172.1	-43.5	7	107	118	129	0.48	0.60	0.71	298.6	94.0	98.0	S_2
14 (late)	172.1	-43.5	7	112	131	150	0.00	0.35	0.79	311.0	91.6	93.3	S_2
41 (early)	172.5	-43.6	6	105	115	125	0.37	0.45	0.53	115.2	91.6	270.0	S_2
41 (late)	172.5	-43.6	6	108	119	129	0.52	0.61	0.72	116.6	108.6	251.9	S_2

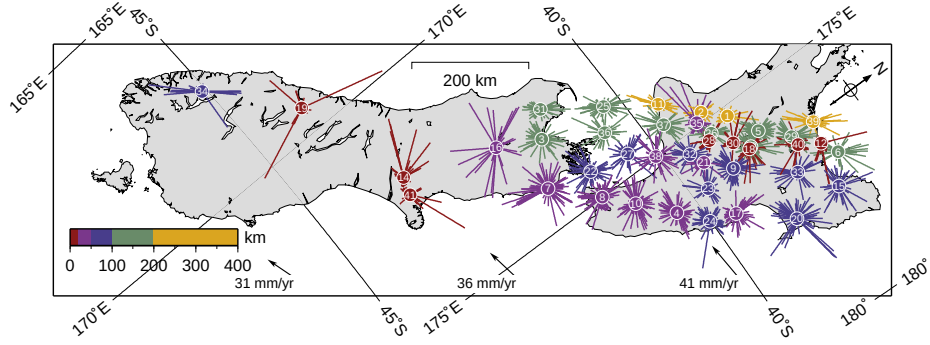


Figure S3: Cluster centroid locations (colored according to depth) for the 41-cluster allocation; the number illustrated at the center of each cluster corresponds to the cluster number in Table S2.

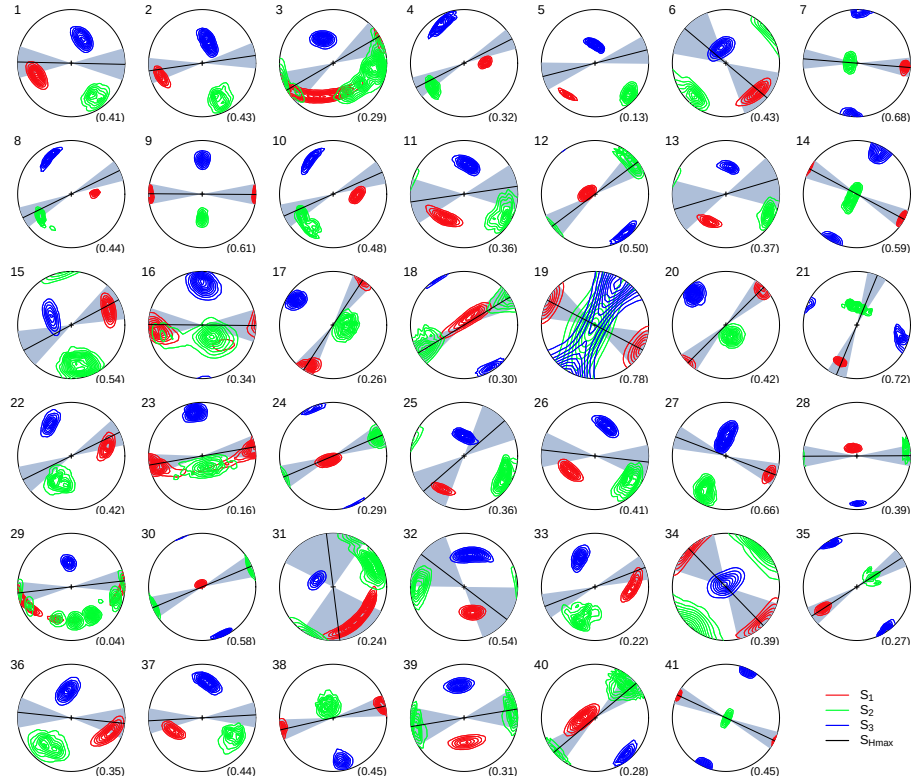


Figure S4: Stereonets showing the geometric distribution of the principal stress axes in each of $k = 41$ clusters (lower-hemisphere projection; north is up, east is to the right). Red, green, and blue contours denote the positions of the S_1 , S_2 , and S_3 axes, and the corresponding direction of S_{Hmax} is marked as a black line with 80% confidence intervals shaded in gray. The cluster number is printed to the upper left of each stereonet, and the posterior mean of the stress ratio v is printed in parentheses to the lower right.

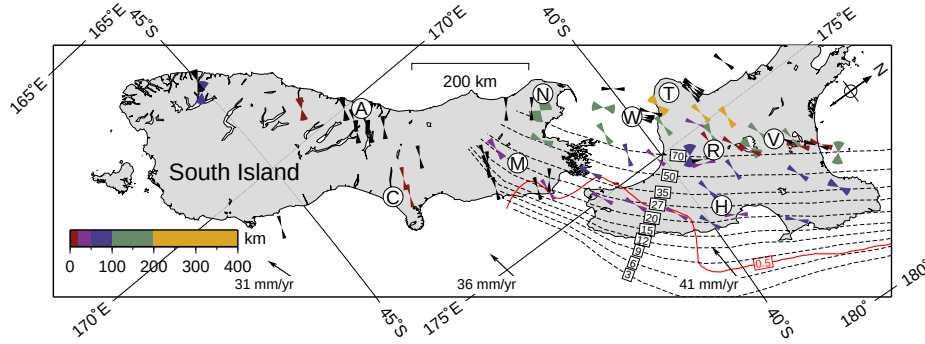


Figure S5: Maps showing the axis of maximum horizontal compression (S_{Hmax}) calculated using the 41-cluster focal mechanism clustering scheme illustrated in Figure S3. The angular extent of the bow-tie symbol illustrates the 80% confidence interval for S_{Hmax} . Also shown (as black bow-tie symbols) are S_{Hmax} orientations obtained by other authors in Fiordland (Reyners et al., 2002), central South Island (Boese et al., 2012), northeast South Island (Balfour et al., 2005), offshore southeastern South Island (Sibson et al., 2011), Taupo Volcanic Zone (Hurst et al., 2002, 2008), Ruapehu area (Hayes et al., 2004), western North Island (Horan, 1997, 2002; Mildren et al., 2001; Mildren and Meyer, 2005). Also illustrated are the $\phi = 0.5$ subduction thrust coupling coefficient contour (red line, points up dip of which are more coupled; Wallace et al., 2012) and depth contours of the Pacific slab (black dashed lines, depths in km; Ansell and Bannister, 1996). A—Alpine Fault; C—Canterbury; H—southern Hawke's Bay; M—Marlborough Fault System; N—northwest Nelson; R—Mt. Ruapehu; T—Mt. Taranaki; V—Taupo Volcanic Zone; W—Wanganui Basin.

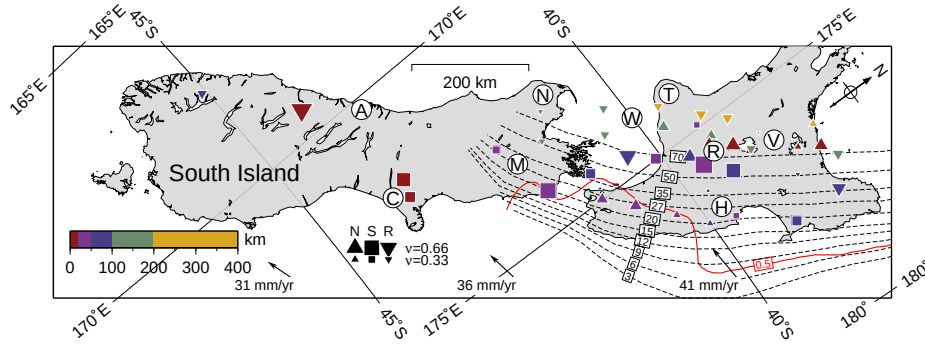


Figure S6: Map of the inferred stress regime $v = (S_1 - S_2)/(S_1 - S_3)$ values calculated using the 41-cluster focal mechanism clustering scheme illustrated in Figure S3. The symbol denotes the stress regime (upright triangle—normal; square—strike-slip; inverted triangle—reverse), the symbol size denotes the value of v , and the symbol color represents depth. $v = 0$ corresponds to an oblate stress ellipsoid ($S_1 = S_2 > S_3$) and $v = 1$ to a prolate stress ellipsoid ($S_1 > S_2 = S_3$). Also illustrated are the $\phi = 0.5$ subduction thrust coupling coefficient contour (red line, points up dip of which are more coupled; Wallace et al., 2012) and depth contours of the Pacific slab (black dashed lines, depths in km; Ansell and Bannister, 1996). A—Alpine Fault; C—Canterbury; H—southern Hawke's Bay; M—Marlborough Fault System; N—northwest Nelson; R—Mt. Ruapehu; T—Mt. Taranaki; V—Taupo Volcanic Zone; W—Wanganui Basin.

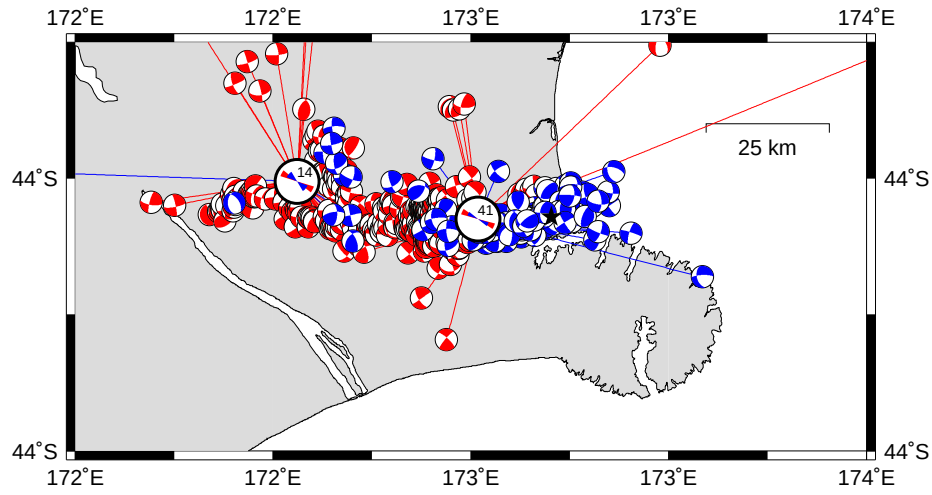


Figure S7: S_{Hmax} directions computed for two clusters in the vicinity of Christchurch using the focal mechanisms of earthquakes occurring before (red) and after (blue) the M_W 6.2 Christchurch earthquake of 22 February 2011. Colored lines link each focal mechanism to the centroid of the cluster it belongs to, and the corresponding S_{Hmax} directions (colored bow-tie symbols) are illustrated at the cluster centroids. For clarity, the extent of the map does not show some of the distant earthquakes: the locations of these events can be seen in Figure S3. The number appearing next to each pair of bow-tie symbols is the cluster number. The black star marks the epicenter of the M_W 6.2 earthquake (after Bannister et al., 2011).